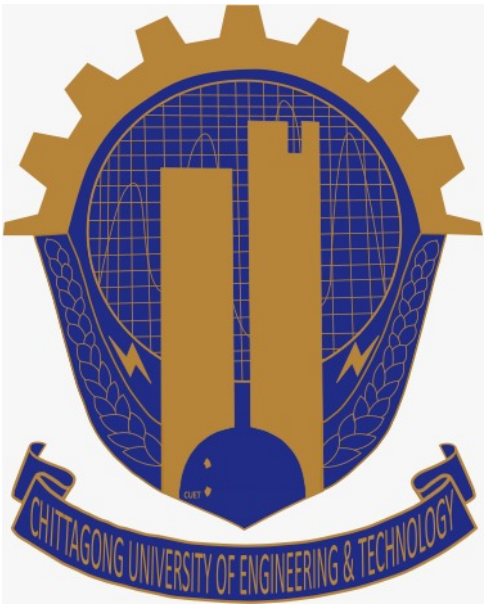
Bachelor of Science in Computer Science & Engineering



StudyTime

Hall Study Room and Library Management System

by

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August, 2026

Project Proposal

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Program	:	B.Sc. Engineering
Tentative Title	:	StudyTime – Hall Study Room and Library Management System

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1 Introduction

StudyTime is a database-driven hall study room management system designed to organize and manage study facilities, seat reservations, library resources, book borrowing, and student complaints.The system provides a centralized platform where student-related information, study-seat information, reservations, library resources, borrowing records, and complaints can be stored and managed efficiently.The project focuses on applying Database Management System concepts in a practical application. The database contains multiple related entities and relationship tables, with primary keys and foreign keys used to maintain data consistency and integrity.StudyTime also provides SQL queries for retrieving useful information from the database, such as student reservations, available seats, reservation history, book information, borrowing history, complaints, and library statistics.

2 Project Overview

StudyTime is developed to simplify the management of study-room and library facilities within a university hall.The system maintains information about students, administrators, halls, seats, reservations, books, authors, borrowing records, complaints, and the relationships between these entities.The database is designed using the relational database model. Different relations are connected using primary keys and foreign keys.The main functionalities represented by the database include:

- Student information management
- Administrator information management
- Study-seat management
- Seat reservation
- Reservation monitoring
- Library management
- Book management
- Book borrowing
- Complaint management
- Administrator monitoring

3 Key Features

The main features of the StudyTime system are as follows:

- **Student Management:** Stores and manages student information including student details and contact information.

- **Administrator Management:** Allows administrators to manage and monitor different activities of the system.
- **Study Seat Management:** Maintains information about study seats, including their availability and other relevant details.
- **Seat Reservation:** Allows students to reserve available study seats and maintains the reservation records in the database.
- **Reservation Monitoring:** Enables administrators to monitor student reservations and manage reservation-related information.
- **Library Management:** Maintains information about libraries and their associated resources.
- **Book Management:** Stores book information and maintains the relationship between books, authors, and libraries.
- **Book Borrowing:** Keeps track of books borrowed by students through borrowing records.
- **Complaint Management:** Allows complaints to be stored and associated with the students who submit them.
- **Phone Number Management:** Maintains separate phone-number information for students and administrators.
- **Database Relationships:** Uses primary keys and foreign keys to establish relationships among different entities and maintain data integrity.
- **SQL Query Support:** Provides SQL queries for retrieving useful information such as reservations, available seats, borrowing records, complaints, and library information.
- **Data Integrity:** Maintains consistency and accuracy of data through relational constraints and properly defined relationships.

4 Technologies Used

The StudyTime project uses database and software development technologies to implement and test the system.

4.1 Database

- MySQL

4.2 Programming and Development Technologies

- Java
- Spring Boot
- React
- JavaScript

4.3 Development Tools

- Visual Studio Code
- MySQL Workbench
- Git
- GitHub

5 System Description

StudyTime contains several entities and relationship tables that together represent the complete database structure. The database stores information related to:

- Students
- Administrators
- Halls
- Seats
- Reservations
- Libraries
- Books
- Authors
- Borrow records
- Complaints

The relationship tables establish connections between students, administrators, seats, reservations, books, and other entities.

6 Entity Relationship Diagram

The initial Entity Relationship (ER) diagram of the StudyTime system was designed to represent the main entities, their attributes, and the relationships among them. It provides an overall representation of the database structure and shows how the different components of the system are connected.

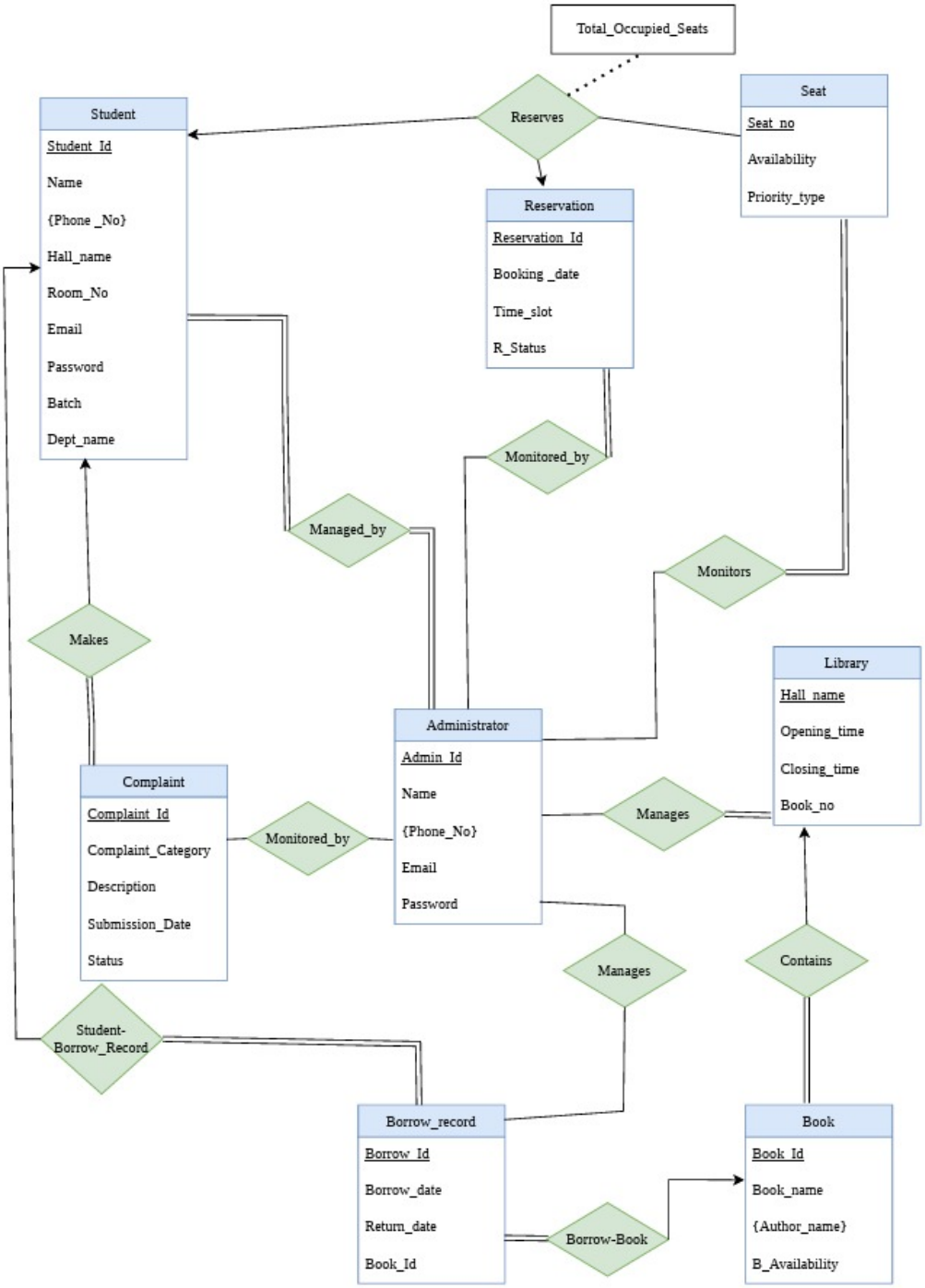


Figure 1: Initial Entity Relationship Diagram of StudyTime

The ER diagram was later revised to improve the database design and to represent

some of the system requirements more clearly. Several changes were made to simplify the relationships and to add important attributes to the entities.

6.1 Changed ER Diagram

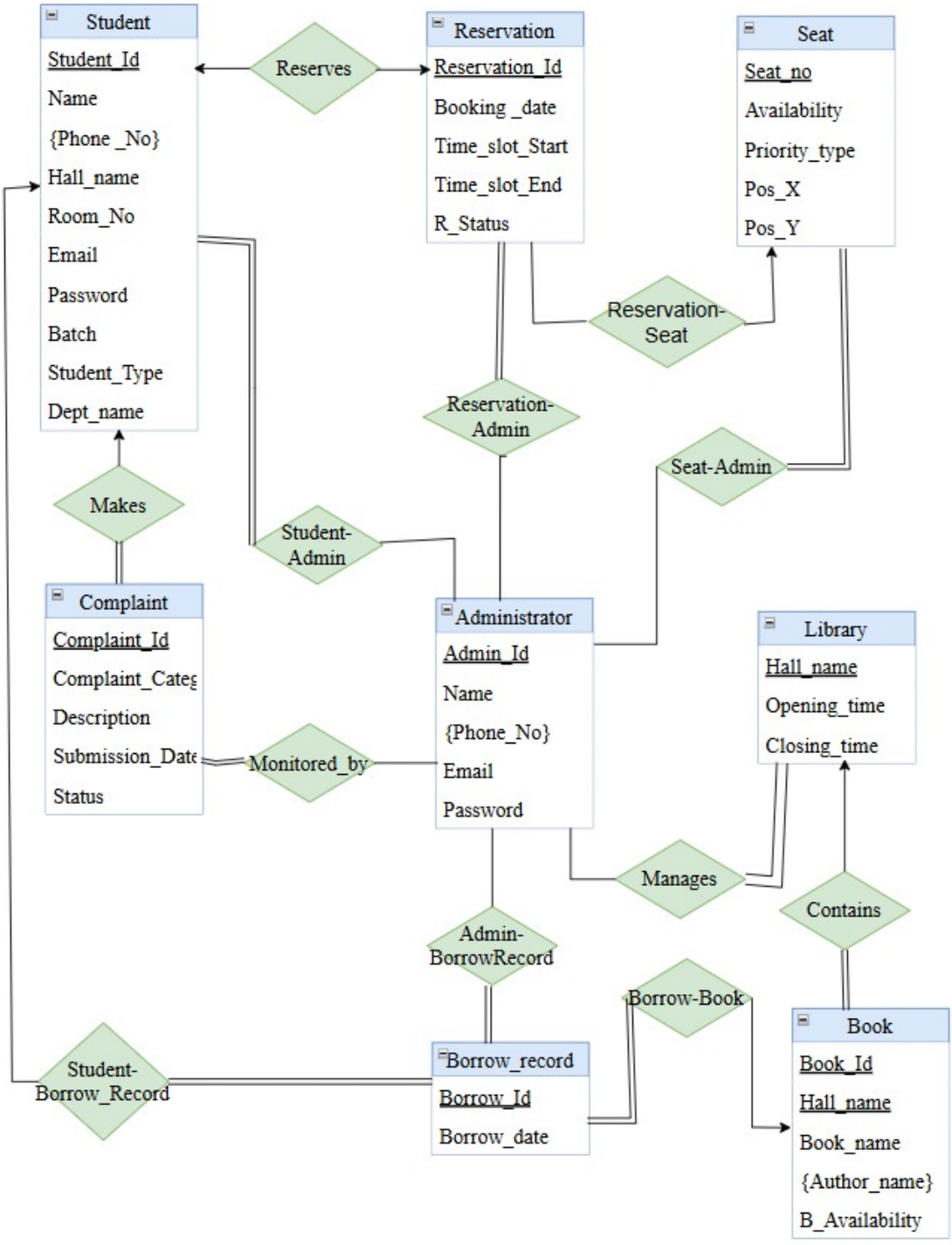


Figure 2: Changed Entity Relationship Diagram of StudyTime

The major changes in the updated ER diagram include converting the ternary relationship into binary relationships to simplify the database structure. The Seat entity was enhanced

by adding Pos_X and Pos_Y attributes to represent the physical position of a seat. The Student entity was updated with a Student Type attribute for student classification. In addition, Hall Name was added to the Book entity to identify the hall associated with a book. These modifications make the ER diagram clearer and more suitable for mapping into the relational database.

7 Relational Mapping

The initial relational mapping of the StudyTime system was created by converting the entities, attributes, and relationships from the Entity Relationship diagram into relational tables. The mapping represents the primary keys, foreign keys, and relationships among the different tables of the system.

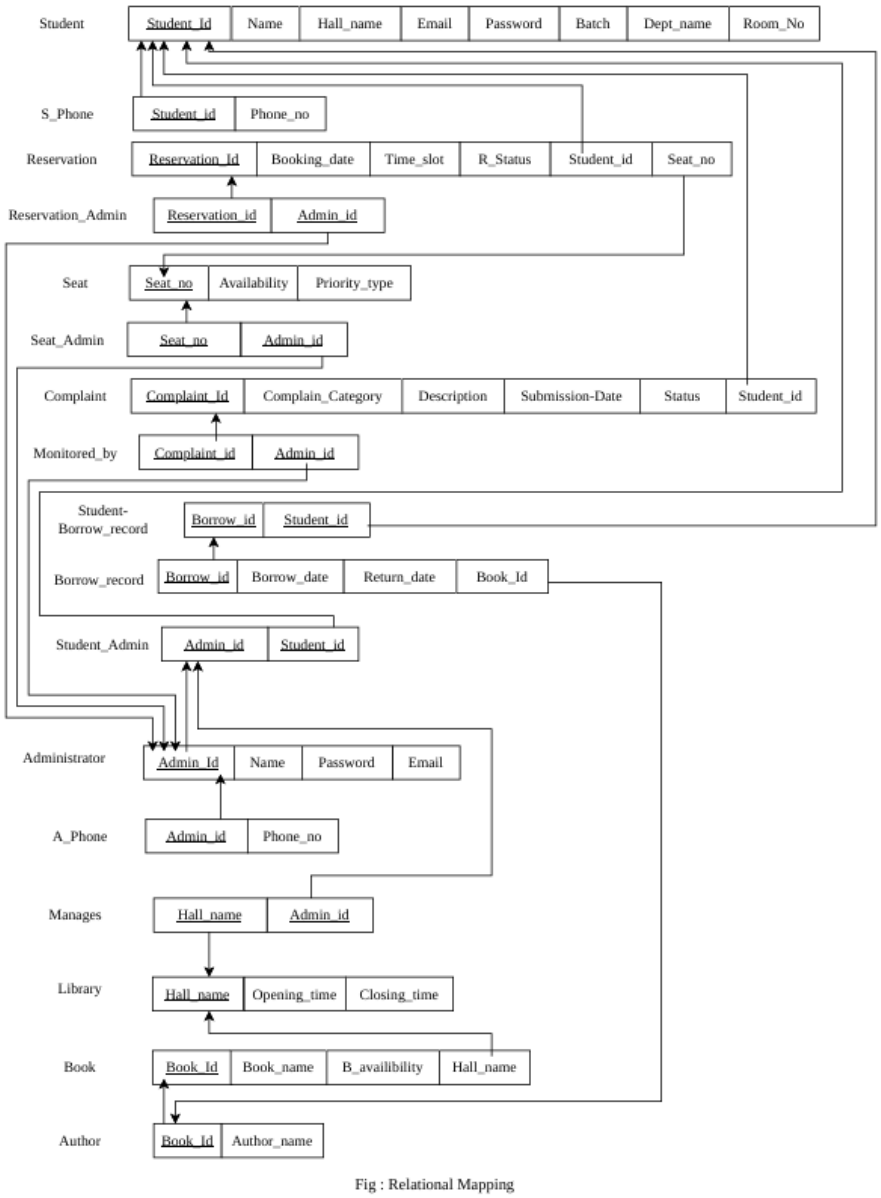


Figure 3: Initial Relational Mapping of StudyTime

The initial relational mapping provided the basic structure of the database, but some issues were identified during the review of the design. In particular, the relationship between the Student and Borrow_record tables was not properly represented, and some primary key constraints were not clearly indicated. The Time_slot attribute was also represented as a single attribute, and the Book relation contained attributes that could be separated for better database organization.

7.1 Changed Relational Mapping

The relational mapping was revised to address these issues and to improve the organization and consistency of the database. The updated mapping contains the necessary foreign keys, decomposed attributes, and separated relations.

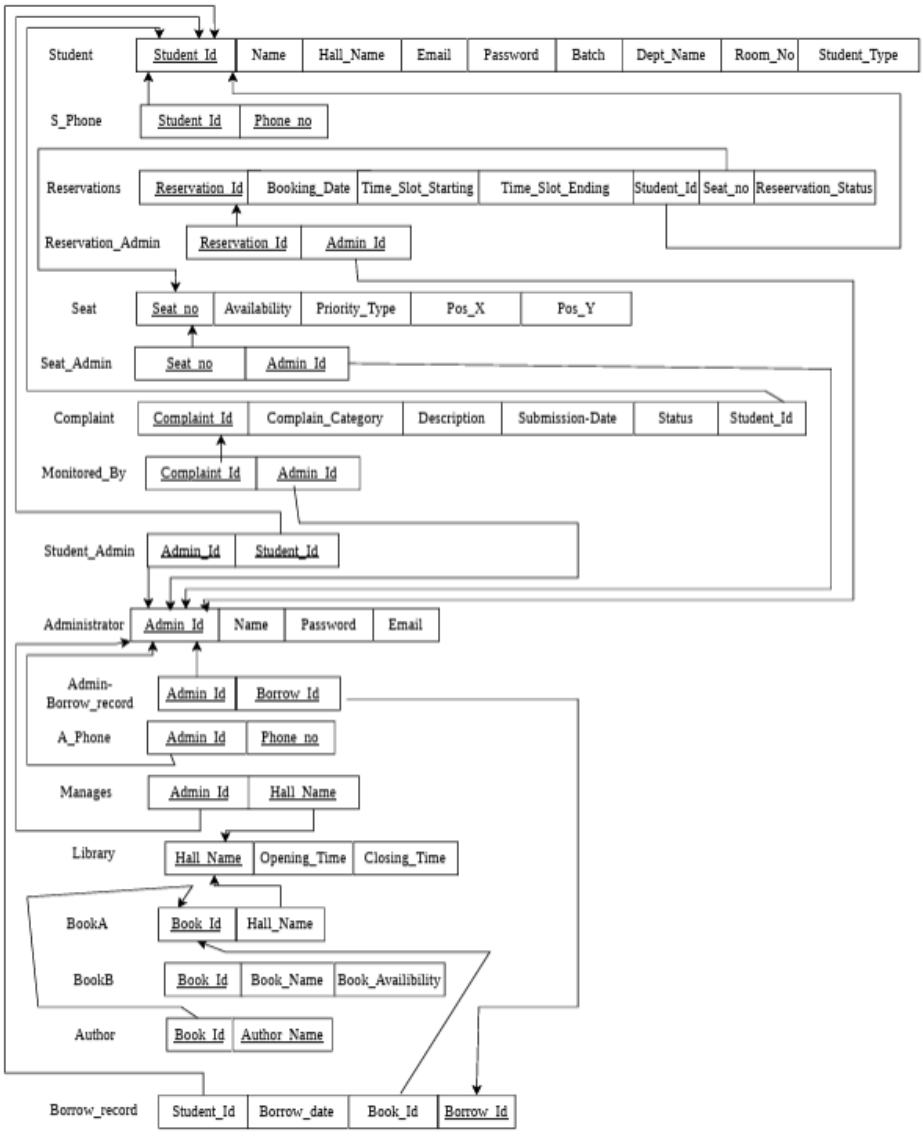


Fig : Relational Mapping

Figure 4: Changed Relational Mapping of StudyTime

The major changes made in the updated relational mapping are described below. First, the Student and Borrow_record tables were properly connected by adding Student_Id to

the Borrow_record table as a foreign key. Second, the primary keys of the relations created from the multi-valued attributes S_Phone, A_Phone, and Author were properly identified and underlined in the updated mapping. Third, for normalization, the Time_slot attribute in the Reservation table was decomposed into Time_Slot_Starting and Time_Slot_Ending. Finally, the Book table was decomposed into two separate relations, BookA and BookB, to organize the book-related information more effectively. These changes improve the integrity, clarity, and normalization of the StudyTime relational database.

8 Database Relations

The following subsections show the relations created for the StudyTime database.

8.1 a_phone Relation

The a_phone relation stores phone numbers associated with administrators.

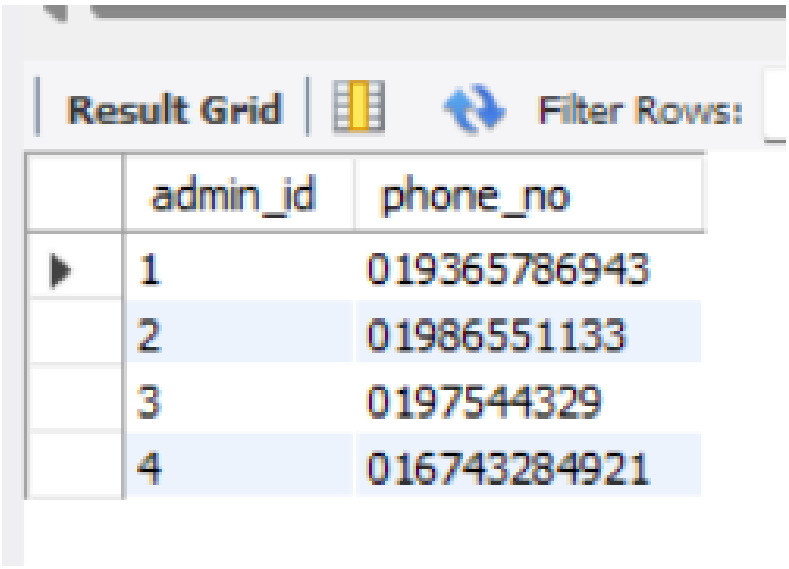


Figure 5: a_phone Relation

8.2 admin_borrow_record Relation

The admin_borrow_record relation represents the relationship between administrators and borrowing records.

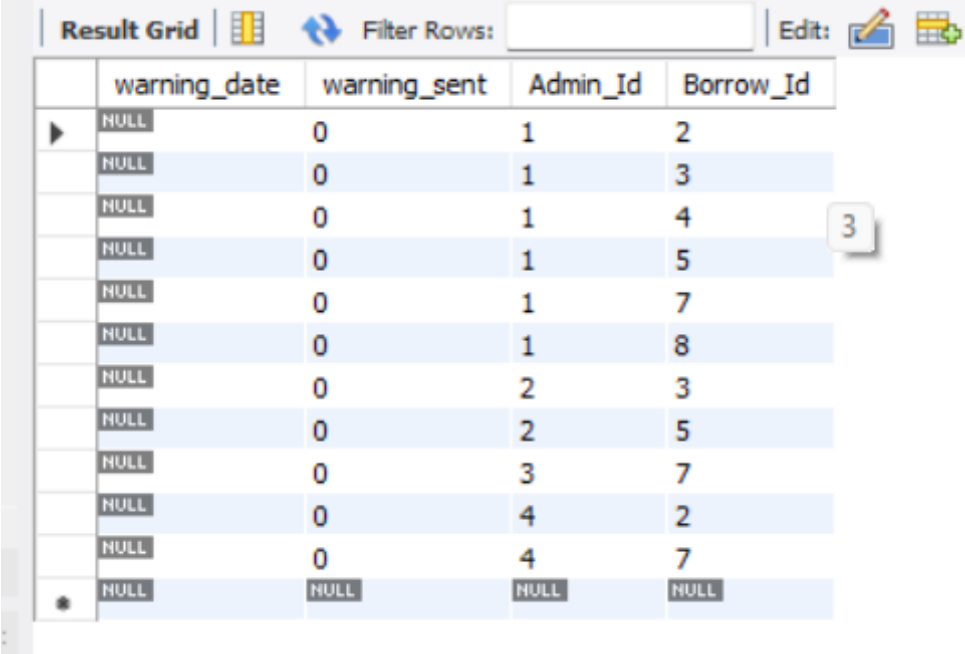


Figure 6: admin_borrow_record Relation

8.3 administrator Relation

The administrator relation stores administrator information.

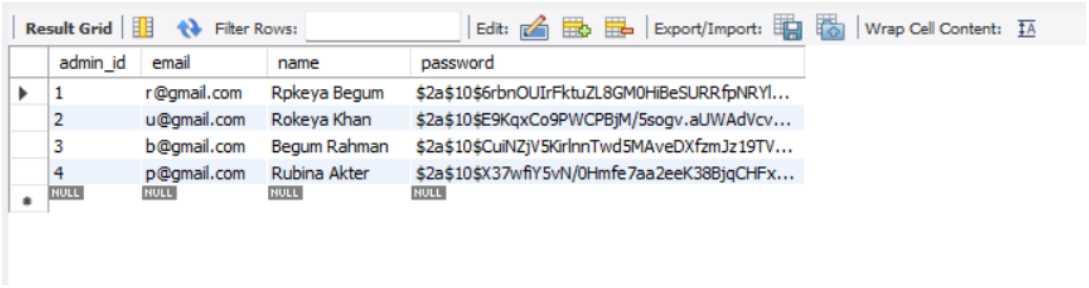


Figure 7: administrator Relation

8.4 author Relation

The author relation stores author information associated with books.

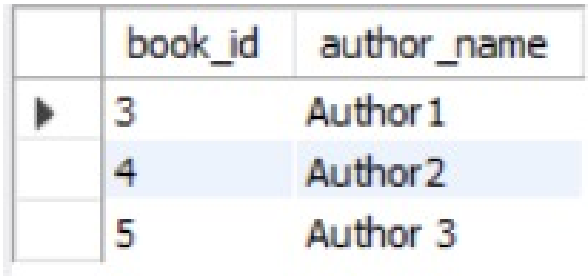
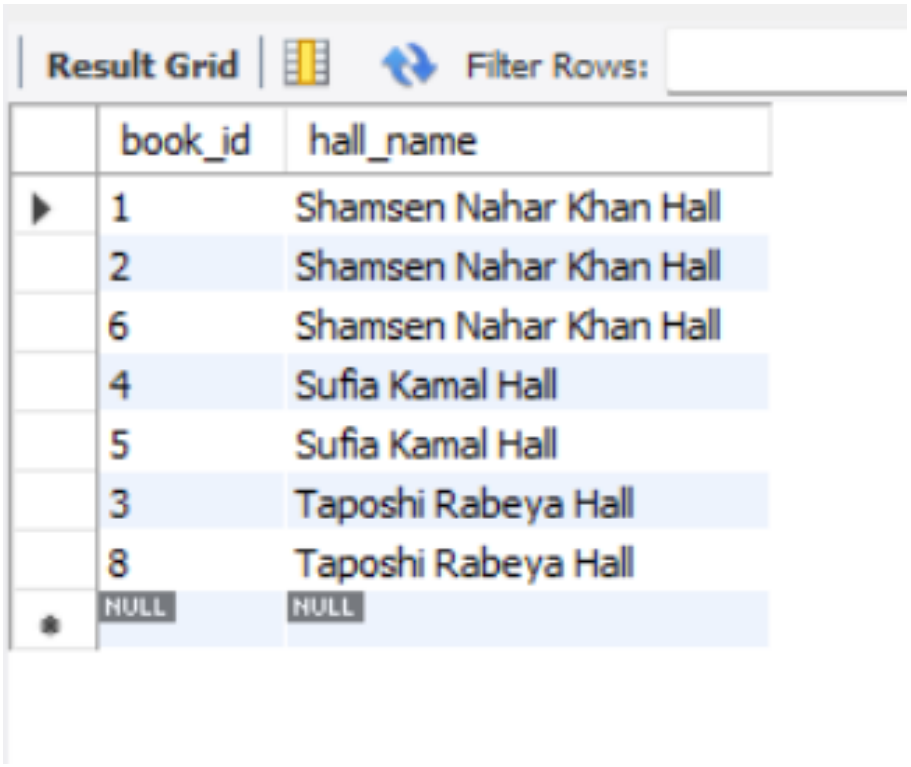


Figure 8: author Relation

8.5 booka Relation

The booka relation stores book and hall-related information.

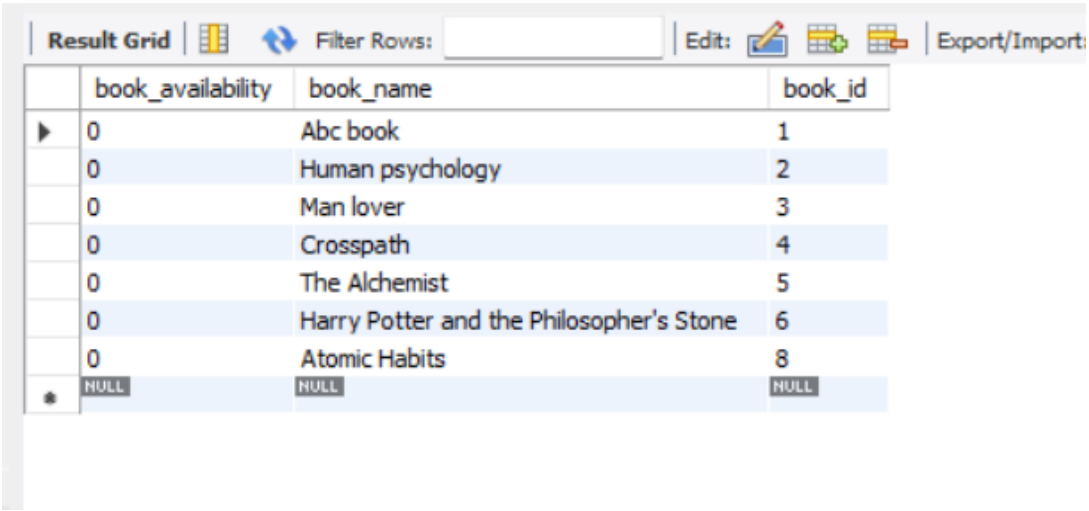


book_id	hall_name
1	Shamsen Nahar Khan Hall
2	Shamsen Nahar Khan Hall
6	Shamsen Nahar Khan Hall
4	Sufia Kamal Hall
5	Sufia Kamal Hall
3	Taposhi Rabeya Hall
8	Taposhi Rabeya Hall
NULL	NULL

Figure 9: booka Relation

8.6 bookb Relation

The bookb relation contains book information and availability data.

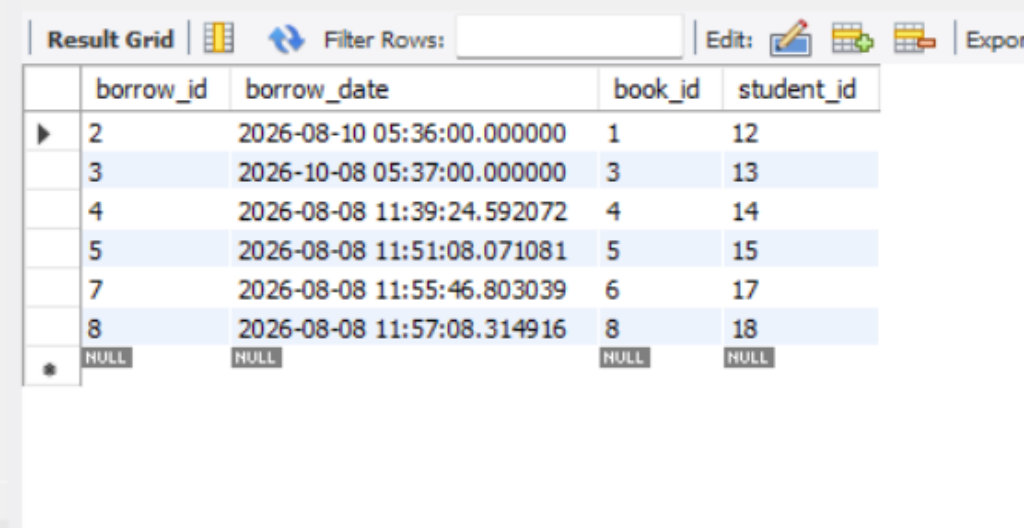


book_availability	book_name	book_id
0	Abc book	1
0	Human psychology	2
0	Man lover	3
0	Crosspath	4
0	The Alchemist	5
0	Harry Potter and the Philosopher's Stone	6
0	Atomic Habits	8
NULL	NULL	NULL

Figure 10: bookb Relation

8.7 borrow_record Relation

The borrow_record relation stores information about book borrowing.

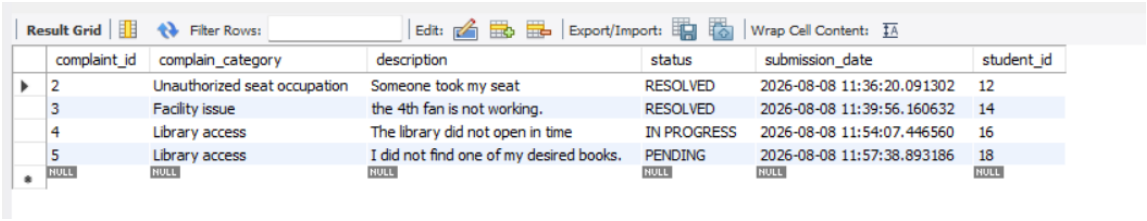


borrow_id	borrow_date	book_id	student_id
2	2026-08-10 05:36:00.000000	1	12
3	2026-10-08 05:37:00.000000	3	13
4	2026-08-08 11:39:24.592072	4	14
5	2026-08-08 11:51:08.071081	5	15
7	2026-08-08 11:55:46.803039	6	17
8	2026-08-08 11:57:08.314916	8	18
NULL	NULL	NULL	NULL

Figure 11: borrow_record Relation

8.8 complaint Relation

The complaint relation stores complaints submitted through the system.

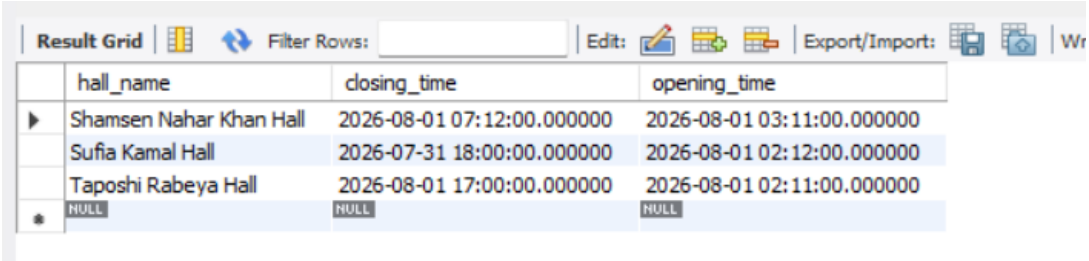


complaint_id	complain_category	description	status	submission_date	student_id
2	Unauthorized seat occupation	Someone took my seat	RESOLVED	2026-08-08 11:36:20.091302	12
3	Facility issue	the 4th fan is not working.	RESOLVED	2026-08-08 11:39:56.160632	14
4	Library access	The library did not open in time	IN PROGRESS	2026-08-08 11:54:07.446560	16
5	Library access	I did not find one of my desired books.	PENDING	2026-08-08 11:57:38.893186	18
NULL	NULL	NULL	NULL	NULL	NULL

Figure 12: complaint Relation

8.9 library Relation

The library relation contains information about the libraries associated with different halls.

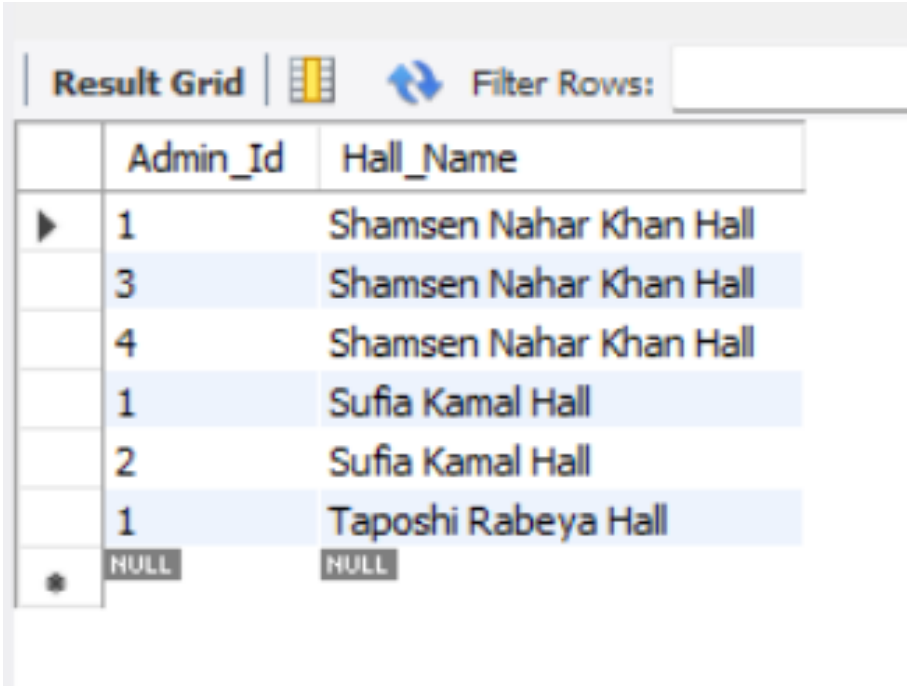


hall_name	closing_time	opening_time
Shamsen Nahar Khan Hall	2026-08-01 07:12:00.000000	2026-08-01 03:11:00.000000
Sufia Kamal Hall	2026-07-31 18:00:00.000000	2026-08-01 02:12:00.000000
Taposhi Rabeya Hall	2026-08-01 17:00:00.000000	2026-08-01 02:11:00.000000
NULL	NULL	NULL

Figure 13: library Relation

8.10 manages Relation

The manages relation represents the management relationship between administrators and halls.

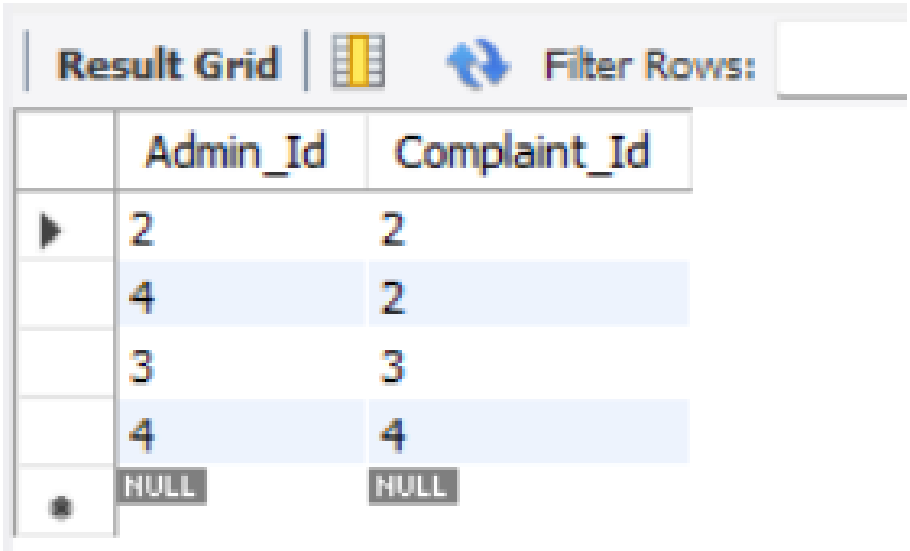


Admin_Id	Hall_Name
1	Shamsen Nahar Khan Hall
3	Shamsen Nahar Khan Hall
4	Shamsen Nahar Khan Hall
1	Sufia Kamal Hall
2	Sufia Kamal Hall
1	Taposhi Rabeya Hall
NULL	NULL

Figure 14: manages Relation

8.11 monitored_by Relation

The monitored_by relation represents the monitoring relationship in the system.



Admin_Id	Complaint_Id
2	2
4	2
3	3
4	4
NULL	NULL

Figure 15: monitored_by Relation

8.12 reservation_admin Relation

The reservation_admin relation associates administrators with reservations.

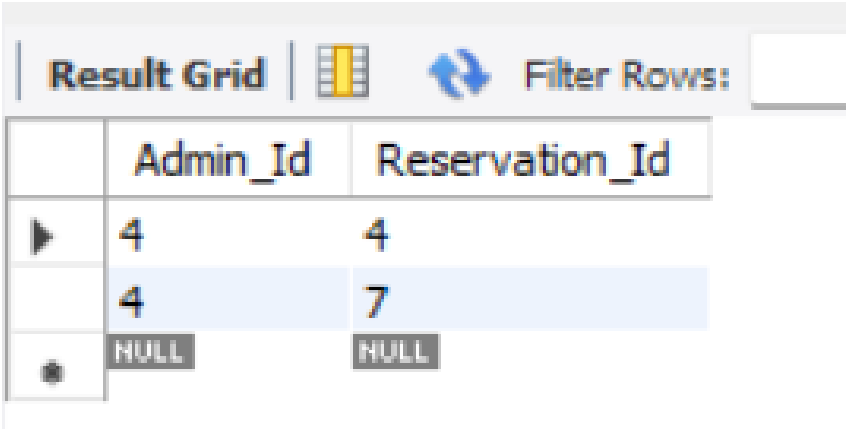


Figure 16: reservation_admin Relation

8.13 reservations Relation

The reservations relation stores study-seat reservation information.

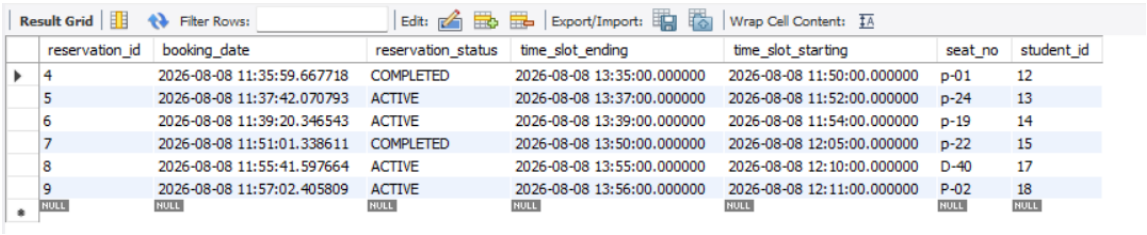


Figure 17: reservations Relation

8.14 s_phone Relation

The s_phone relation stores phone numbers associated with students.

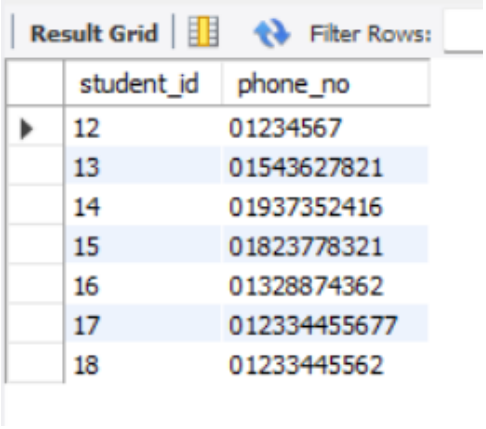
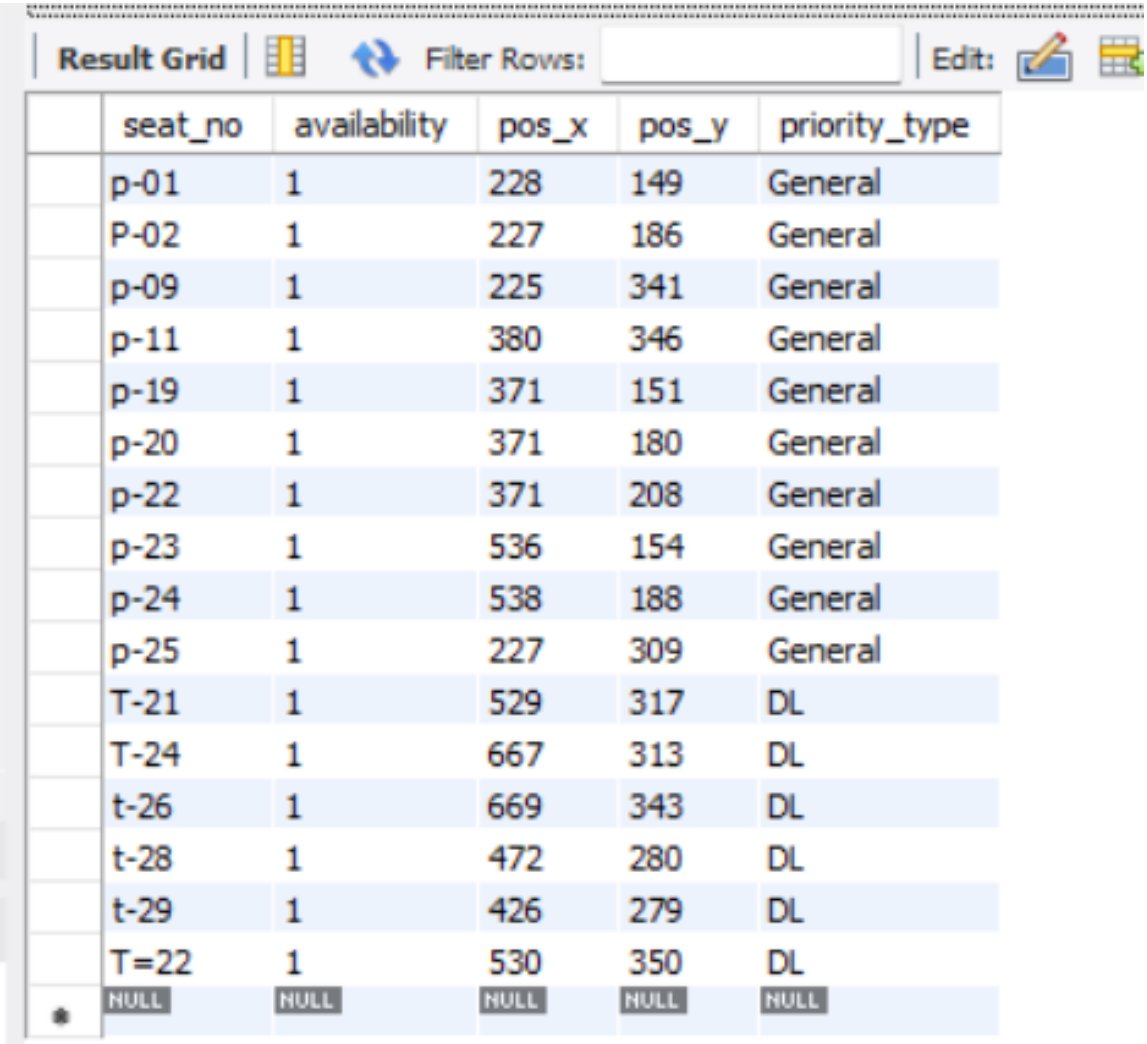


Figure 18: s_phone Relation

8.15 seat Relation

The seat relation stores study-seat information including availability, position, and priority type.



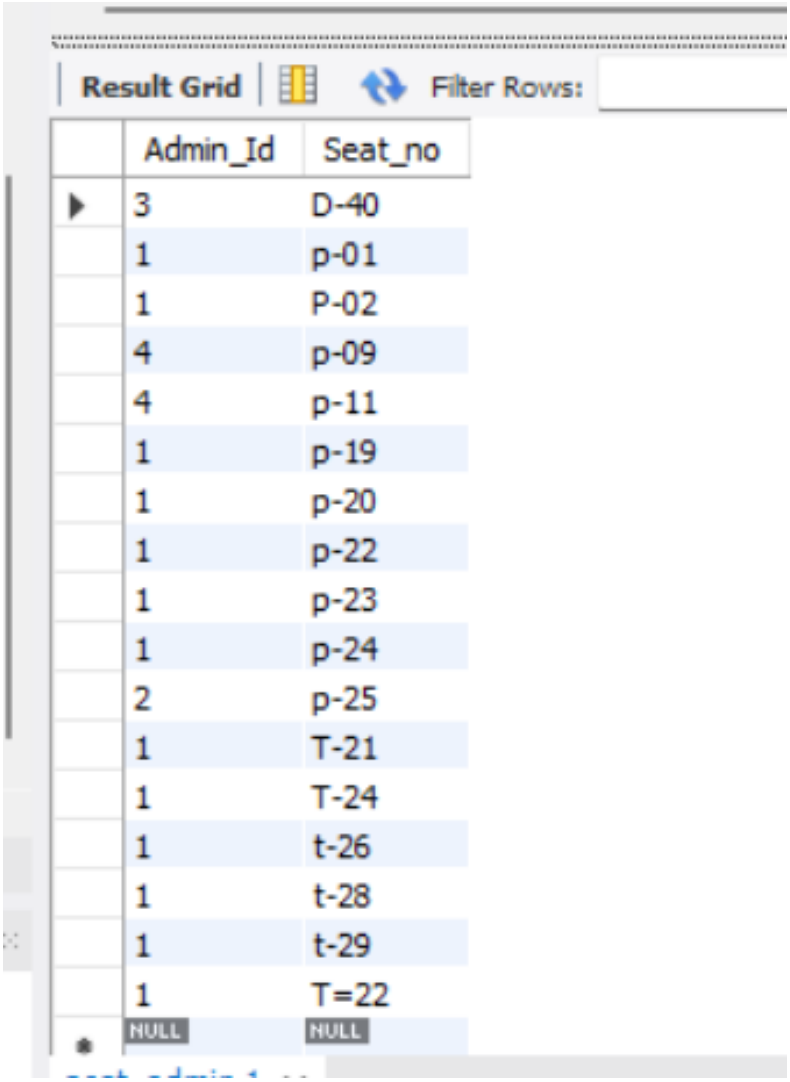
Result Grid | Filter Rows: | Edit:

seat_no	availability	pos_x	pos_y	priority_type
p-01	1	228	149	General
P-02	1	227	186	General
p-09	1	225	341	General
p-11	1	380	346	General
p-19	1	371	151	General
p-20	1	371	180	General
p-22	1	371	208	General
p-23	1	536	154	General
p-24	1	538	188	General
p-25	1	227	309	General
T-21	1	529	317	DL
T-24	1	667	313	DL
t-26	1	669	343	DL
t-28	1	472	280	DL
t-29	1	426	279	DL
T=22	1	530	350	DL
NULL	NULL	NULL	NULL	NULL

Figure 19: seat Relation

8.16 seat_admin Relation

The seat_admin relation represents the relationship between administrators and seats.



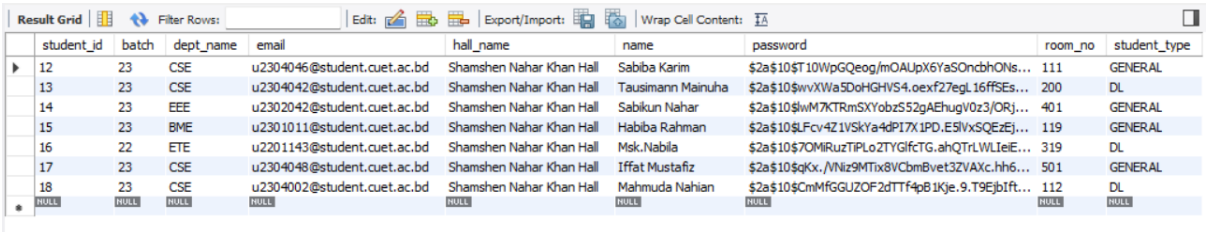
Result Grid | Filter Rows:

Admin_Id	Seat_no
3	D-40
1	p-01
1	P-02
4	p-09
4	p-11
1	p-19
1	p-20
1	p-22
1	p-23
1	p-24
2	p-25
1	T-21
1	T-24
1	t-26
1	t-28
1	t-29
1	T=22
NULL	NULL

Figure 20: seat_admin Relation

8.17 student Relation

The student relation stores student information such as student ID, name, department, hall, room number, email, and other attributes.



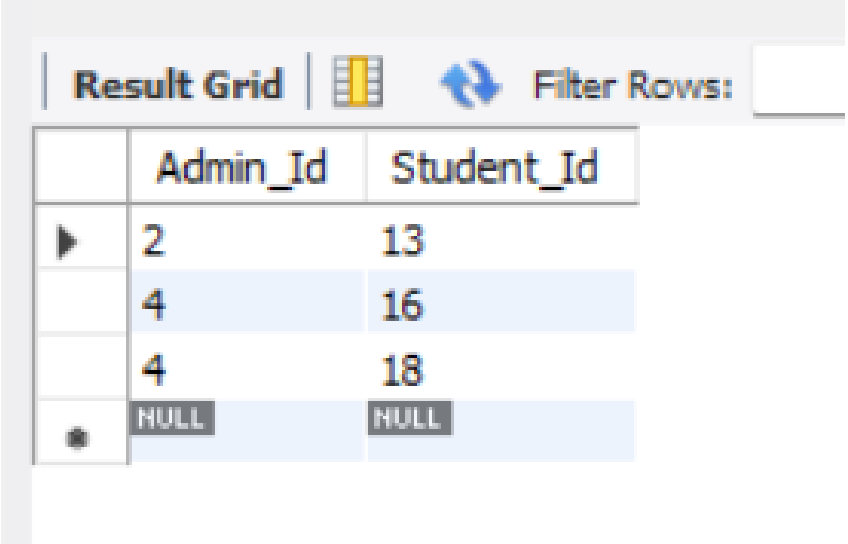
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

student_id	batch	dept_name	email	hall_name	name	password	room_no	student_type
12	23	CSE	u2304046@student.cuet.ac.bd	Shamshen Nahar Khan Hall	Sabiba Karim	$2a$10$T10WpGQeog/mOAUpX6YaSOncbhONs...	111	GENERAL
13	23	CSE	u2304042@student.cuet.ac.bd	Shamshen Nahar Khan Hall	Tausimann Mainuha	$2a$10$wvXWa5DoHGHVS4.oexf27egL16ffSEs...	200	DL
14	23	EEE	u2302042@student.cuet.ac.bd	Shamshen Nahar Khan Hall	Sabikun Nahar	$2a$10$lwM7KTRmSXYobzS52gAEhugV0z3/ORj...	401	GENERAL
15	23	BME	u2301011@student.cuet.ac.bd	Shamshen Nahar Khan Hall	Habiba Rahman	$2a$10$LFcv4Z1VSkYa4dPI7X1PD.ESlVxSQEzEj...	119	GENERAL
16	22	ETE	u2201143@student.cuet.ac.bd	Shamshen Nahar Khan Hall	Msk.Nabila	$2a$10$7OMiRuzTiPLoZTYGlfcTG.ahQTrLWLIeiE...	319	DL
17	23	CSE	u2304048@student.cuet.ac.bd	Shamshen Nahar Khan Hall	Iffat Mustafiz	$2a$10$qKx./VNiz9MTix8VCbmBvet3ZVAXc.hh6...	501	GENERAL
18	23	CSE	u2304002@student.cuet.ac.bd	Shamshen Nahar Khan Hall	Mahmuda Nahian	$2a$10$CmMfGGUZOF2dTTf4pB1Kje.9.T9EjbIft...	112	DL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 21: student Relation

8.18 student_admin Relation

The student_admin relation represents the relationship between students and administrators.



	Admin_Id	Student_Id
▶	2	13
	4	16
	4	18
*	NULL	NULL

Figure 22: student_admin Relation

9 SQL Queries

The following SQL queries were implemented to retrieve useful information from the StudyTime database.

9.1 Query 1 – Display Students and Their Reservations

This query displays the student ID, student name, and reservation ID of students who have made reservations.

```
1 •  SELECT s.student_id,
2           s.name,
3           r.reservation_id
4    FROM student s
5    JOIN reservations r
6    ON s.student_id = r.student_id;
7
```

Figure 23: Query for Displaying Students and Their Reservations

Result Grid | Filter Rows: | Export:

student_id	name	reservation_id
12	Sabiba Karim	4
13	Tausimann Mainuha	5
14	Sabikun Nahar	6
15	Habiba Rahman	7
17	Iffat Mustafiz	8
18	Mahmuda Nahian	9

Figure 24: Output of Query 1

9.2 Query 2 – Display Available Seats

This query retrieves all seats whose availability value is 1.

```
SELECT *
FROM Seat
WHERE availability=1;
```

Figure 25: SQL Query for Displaying Available Seats

Result Grid | Filter Rows: | Edit:

seat_no	availability	pos_x	pos_y	priority_type
p-01	1	228	149	General
P-02	1	227	186	General
p-09	1	225	341	General
p-11	1	380	346	General
p-19	1	371	151	General
p-20	1	371	180	General
p-22	1	371	208	General
p-23	1	536	154	General
p-24	1	538	188	General
p-25	1	227	309	General
T-21	1	529	317	DL
T-24	1	667	313	DL
t-26	1	669	343	DL
t-28	1	472	280	DL
t-29	1	426	279	DL
T=22	1	530	350	DL
NULL	NULL	NULL	NULL	NULL

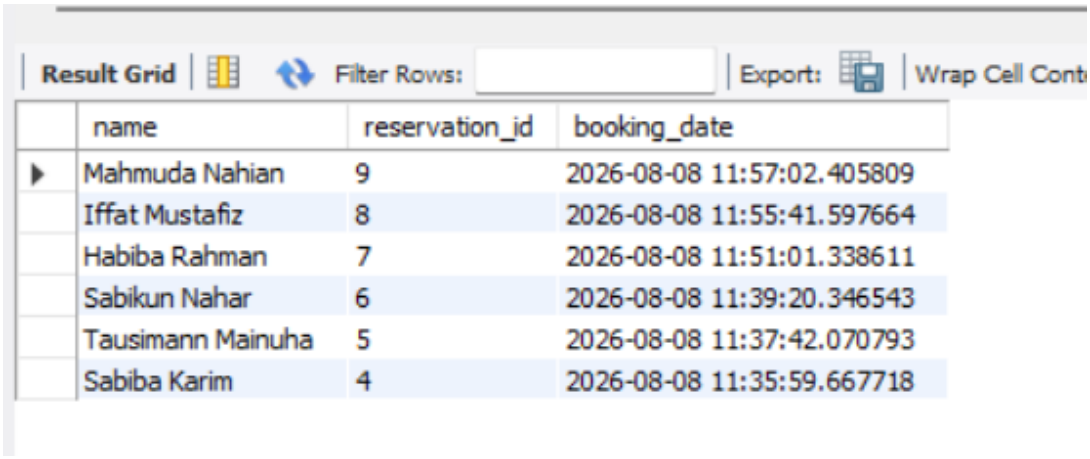
Figure 26: Output of Query 2

9.3 Query 3 – Display Student Reservation History

The query retrieves student names, reservation IDs, and booking dates and orders the records according to booking date.

```
1    SELECT s.name,
2           r.reservation_id,
3           r.booking_date
4    FROM student s
5    JOIN reservations r
6    ON s.student_id = r.student_id
7    ORDER BY r.booking_date DESC;
```

Figure 27: SQL Query for Student Reservation History



name	reservation_id	booking_date
Mahmuda Nahian	9	2026-08-08 11:57:02.405809
Iffat Mustafiz	8	2026-08-08 11:55:41.597664
Habiba Rahman	7	2026-08-08 11:51:01.338611
Sabikun Nahar	6	2026-08-08 11:39:20.346543
Tausimann Mainuha	5	2026-08-08 11:37:42.070793
Sabiba Karim	4	2026-08-08 11:35:59.667718

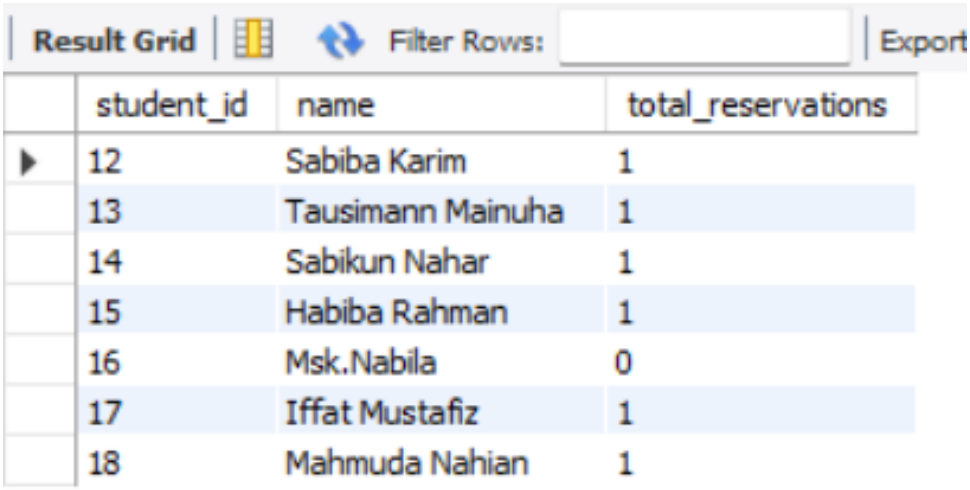
Figure 28: Output of Query 3

9.4 Query 4 – Count Reservations per Student

This query counts the total number of reservations made by each student.

```sql
SELECT s.student_id,
       s.name,
       COUNT(r.reservation_id) AS total_reservations
FROM student s
LEFT JOIN reservations r
ON s.student_id = r.student_id
GROUP BY s.student_id, s.name;
```

Figure 29: SQL Query for Counting Reservations per Student



student_id	name	total_reservations
12	Sabiba Karim	1
13	Tausimann Mainuha	1
14	Sabikun Nahar	1
15	Habiba Rahman	1
16	Msk.Nabila	0
17	Iffat Mustafiz	1
18	Mahmuda Nahian	1

Figure 30: Output of Query 4

9.5 Query 5 – Display Books in the Library

This query combines book-related relations to display book IDs, book names, hall names, and book availability.

```
1    SELECT a.book_id,
2           b.book_name,
3           a.hall_name,
4           b.book_availability
5    FROM booka a
6    JOIN bookb b
7    ON a.book_id = b.book_id;
```

Figure 31: SQL Query for Displaying Books in the Library

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

book_id	book_name	hall_name	book_availability
1	Abc book	Shamsen Nahar Khan Hall	0
2	Human psychology	Shamsen Nahar Khan Hall	0
6	Harry Potter and the Philosopher's Stone	Shamsen Nahar Khan Hall	0
4	Crosspath	Sufia Kamal Hall	0
5	The Alchemist	Sufia Kamal Hall	0
3	Man lover	Taposhi Rabeya Hall	0
8	Atomic Habits	Taposhi Rabeya Hall	0

Figure 32: Output of Query 5

9.6 Query 6 – Display Borrowing History

This query displays the student name, book name, and borrowing date for borrowing records.

```sql
SELECT s.name,
       b.book_name,
       br.borrow_date
FROM borrow_record br
JOIN student s
ON br.student_id = s.student_id
JOIN bookb b
ON br.book_id = b.book_id
ORDER BY br.borrow_date DESC;
```

Figure 33: SQL Query for Borrowing History

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	book_name	borrow_date
Tausimann Mainuha	Man lover	2026-10-08 05:37:00.000000
Sabiba Karim	Abc book	2026-08-10 05:36:00.000000
Mahmuda Nahian	Atomic Habits	2026-08-08 11:57:08.314916
Iffat Mustafiz	Harry Potter and the Philosopher's Stone	2026-08-08 11:55:46.803039
Habiba Rahman	The Alchemist	2026-08-08 11:51:08.071081
Sabikun Nahar	Crosspath	2026-08-08 11:39:24.592072

Figure 34: Output of Query 6

9.7 Query 7 – Display Students With Active Borrowings

This query identifies borrowing records based on book availability and displays the relevant student and book information.

```sql
SELECT s.student_id,
       b.book_name,
       br.borrow_date
FROM Borrow_Record br
JOIN Student s
ON br.student_id = s.student_id
JOIN BookB b
ON br.book_id = b.book_id
WHERE b.book_availability = 0;
```

Figure 35: SQL Query for Students With Active Borrowings

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	book_name	borrow_date
12	Abc book	2026-08-10 05:36:00.000000
13	Man lover	2026-10-08 05:37:00.000000
14	Crosspath	2026-08-08 11:39:24.592072
15	The Alchemist	2026-08-08 11:51:08.071081
17	Harry Potter and the Philosopher's Stone	2026-08-08 11:55:46.803039
18	Atomic Habits	2026-08-08 11:57:08.314916

Figure 36: Output of Query 7

9.8 Query 8 – Display Complaints and Students

This query displays complaint information along with the name of the student who submitted each complaint.

```
1  SELECT s.name,
2         c.complaint_id,
3         c.description,
4         c.status
5  FROM complaint c
6  JOIN student s
7  ON c.student_id = s.student_id
8  ORDER BY c.complaint_id DESC;
```

Figure 37: SQL Query for Complaints and Students

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	complaint_id	description	status
Mahmuda Nahian	5	I did not find one of my desired books.	PENDING
Msk.Nabila	4	The library did not open in time	IN PROGRESS
Sabikun Nahar	3	the 4th fan is not working.	RESOLVED
Sabiba Karim	2	Someone took my seat	RESOLVED

Figure 38: Output of Query 8

9.9 Query 9 – Count Books in Each Library

This query calculates the number of books available in each hall library.

```sql
SELECT l.hall_name,
       COUNT(b.book_id) AS total_books
FROM Library l
LEFT JOIN BookA b
ON l.hall_name = b.hall_name
GROUP BY l.hall_name;
```

Figure 39: SQL Query for Counting Books in Each Library

Result Grid | Filter Rows: | Expo

hall_name	total_books
Shamsen Nahar Khan Hall	3
Sufia Kamal Hall	2
Taposhi Rabeya Hall	2

Figure 40: Output of Query 9

9.10 Query 10 – Display All Reservations With Seat Information

This query combines reservation and seat information to display reservation details together with seat availability and positional information.

```sql
SELECT r.reservation_id,
       r.booking_date,
       r.reservation_status,
       r.time_slot_starting,
       r.time_slot_ending,
       r.seat_no,
       s.availability,
       s.pos_x,
       s.pos_y,
       s.priority_type
FROM Reservations r
JOIN Seat s
ON r.seat_no = s.seat_no;
```

Figure 41: SQL Query for Reservations With Seat Information

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

reservation_id	booking_date	reservation_status	time_slot_starting	time_slot_ending	seat_no	availability	pos_x	pos_y	priority_type
4	2026-08-08 11:35:59.667718	COMPLETED	2026-08-08 11:50:00.000000	2026-08-08 13:35:00.000000	p-01	1	228	149	General
5	2026-08-08 11:37:42.070793	ACTIVE	2026-08-08 11:52:00.000000	2026-08-08 13:37:00.000000	p-24	1	538	188	General
6	2026-08-08 11:39:20.346543	ACTIVE	2026-08-08 11:54:00.000000	2026-08-08 13:39:00.000000	p-19	1	371	151	General
7	2026-08-08 11:51:01.338611	COMPLETED	2026-08-08 12:05:00.000000	2026-08-08 13:50:00.000000	p-22	1	371	208	General
8	2026-08-08 11:55:41.597664	ACTIVE	2026-08-08 12:10:00.000000	2026-08-08 13:55:00.000000	D-40	1	379	310	DL
9	2026-08-08 11:57:02.405809	ACTIVE	2026-08-08 12:11:00.000000	2026-08-08 13:56:00.000000	P-02	1	227	186	General

Figure 42: Output of Query 10

10 System Implementation

The StudyTime system was implemented as a web-based application that provides different functionalities for students and administrators. The system includes user authentication, registration, seat management, reservation management, library and book management, complaint management, and administrative operations. The following sections present the main implemented pages and functionalities of the StudyTime system.

10.1 Authentication

The authentication module provides separate login and registration functionalities for students and administrators. Users can create an account and log into the system according to their respective roles.

10.1.1 Student Login

The Student Login page allows registered students to access the StudyTime system using their login credentials. After successful authentication, the student is directed to the student dashboard and can access the functionalities available to students.

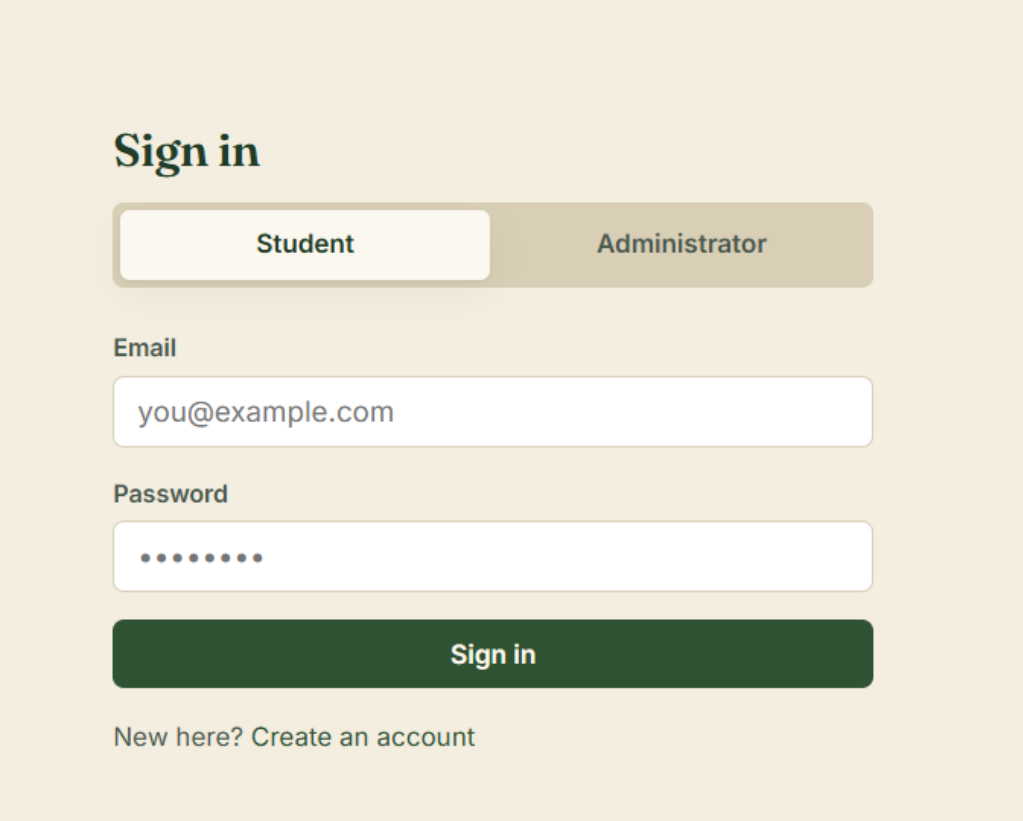


Figure 43: Student Login Page

10.1.2 Admin Login

The Admin Login page provides a separate authentication interface for administrators. After successful login, administrators can access the administrative functionalities of the StudyTime system.

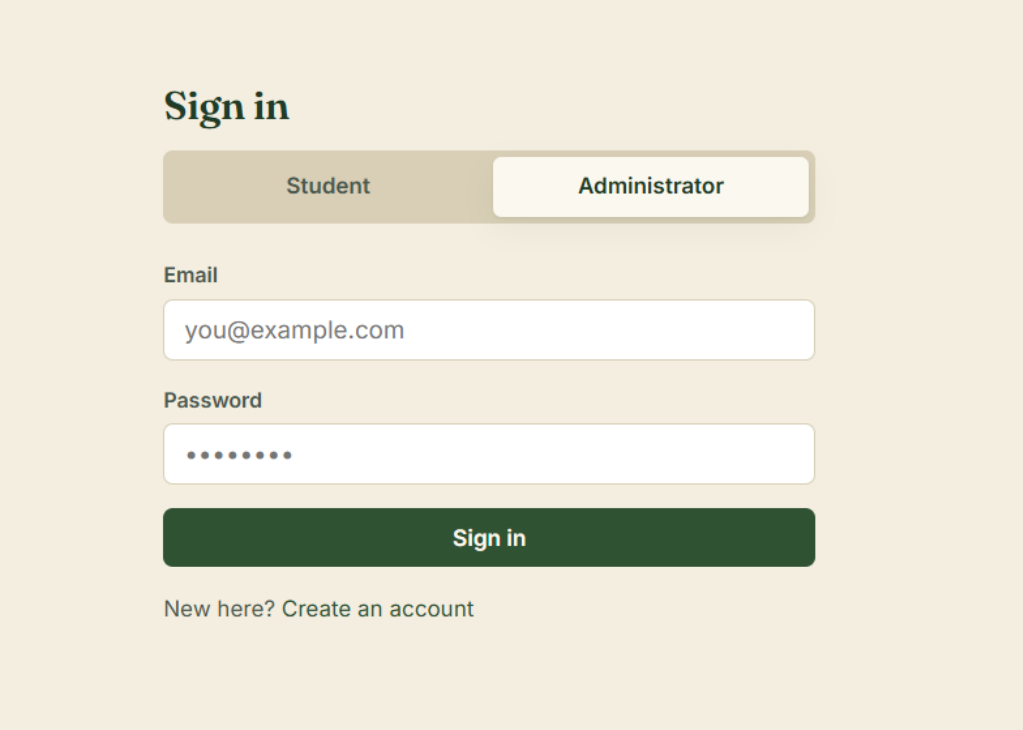


Figure 44: Admin Login Page

10.1.3 Student Registration

The Student Registration page allows new students to create an account in the StudyTime system by providing the required registration information.

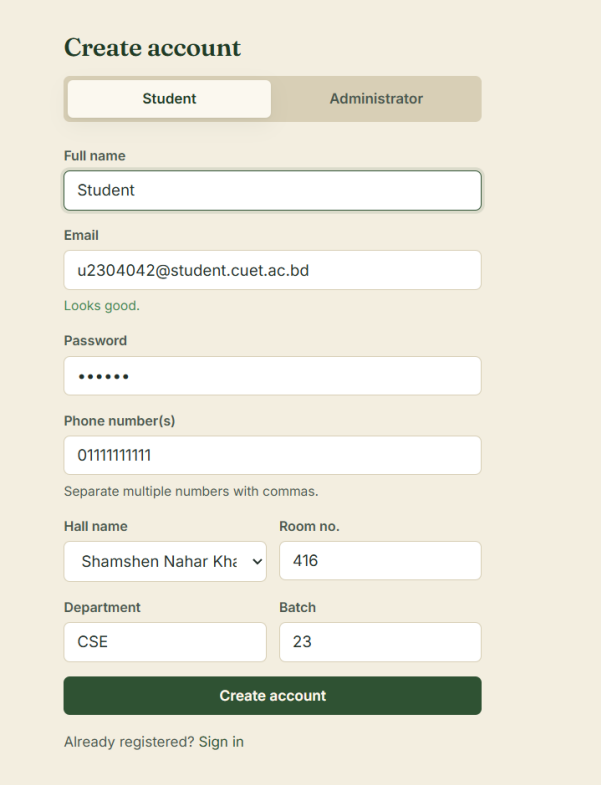


Figure 45: Student Registration Page

10.1.4 Admin Registration

The Admin Registration page allows authorized administrators to create an administrator account and access the administrative features of the system.

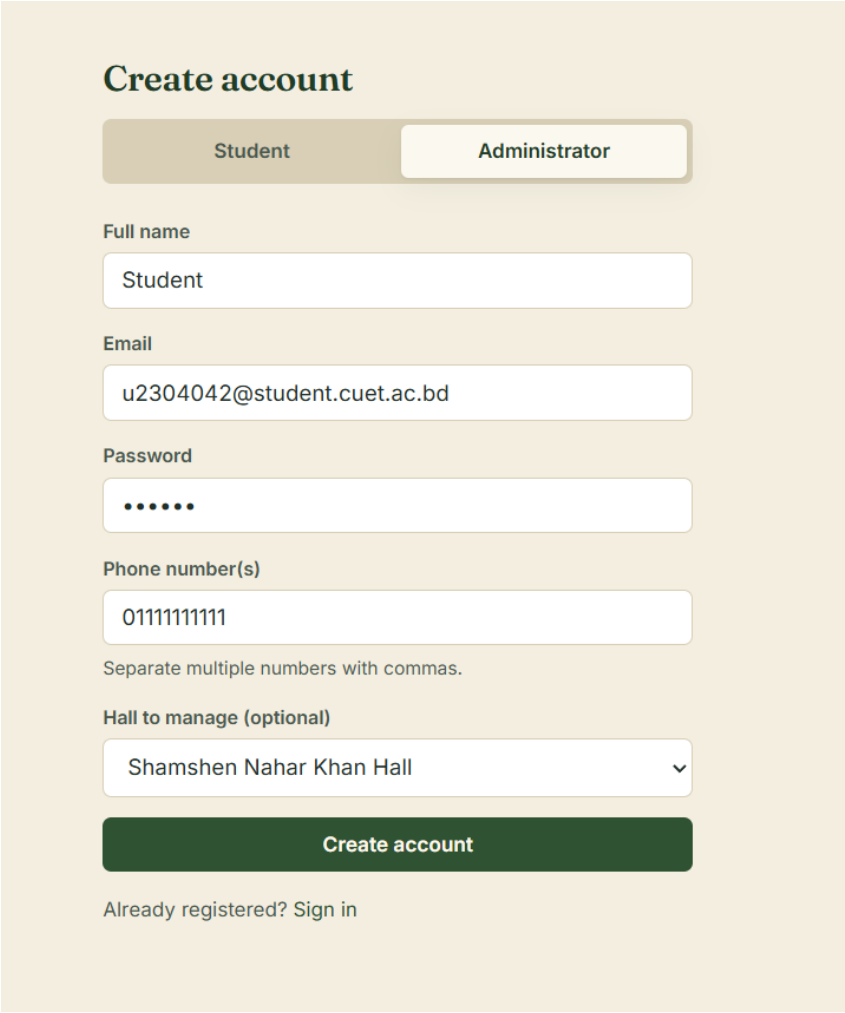


Figure 46: Admin Registration Page

10.2 Dashboard

The dashboard provides an overview of the StudyTime system and gives users access to the major functionalities available according to their role. Students can navigate to seat reservations, library resources, borrowing, and complaints, while administrators can access system management functionalities.

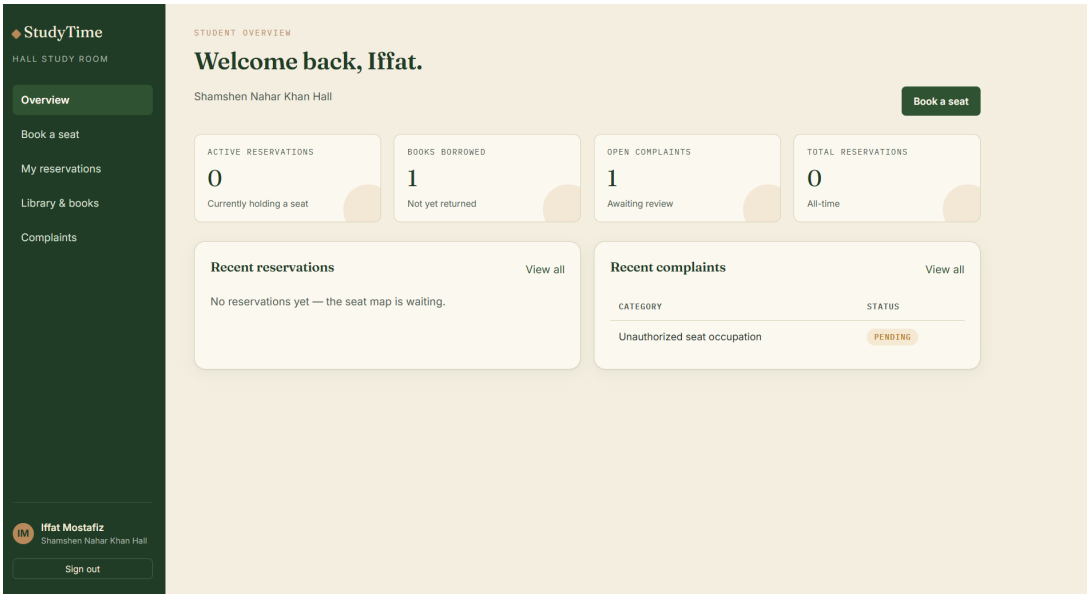


Figure 47: StudyTime Dashboard

10.3 Student and Administrator Functionalities

The main functional modules of StudyTime are available through the student and administrator interfaces. The following sections show the implemented seat management, reservation, library, and complaint functionalities from both sides of the system.

10.3.1 Seat Map

The Seat Map provides a visual representation of the available study seats. Students can view the seating arrangement and identify seats that can be reserved. The administrator side provides the corresponding functionality for monitoring and managing the seats.

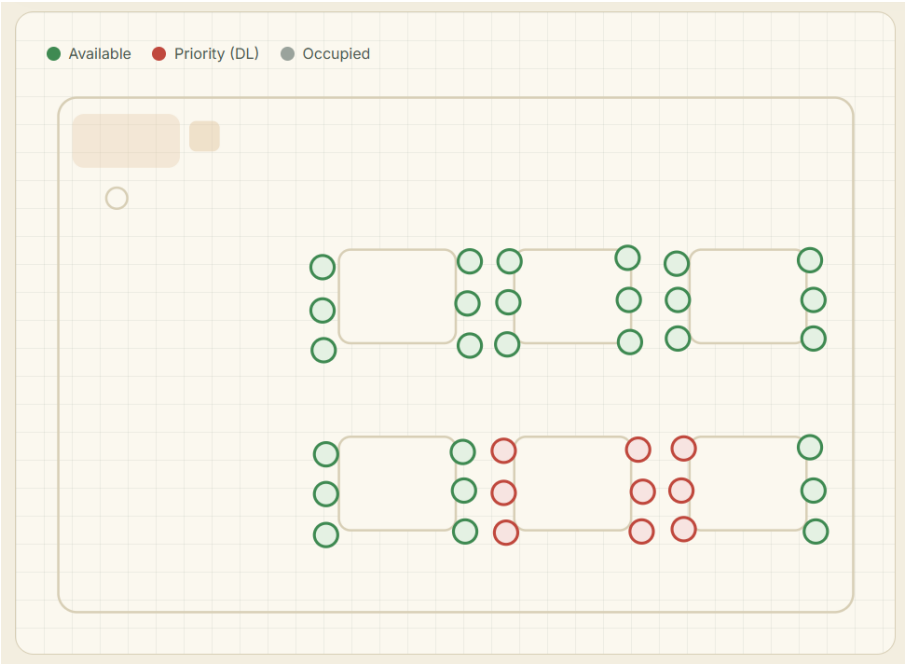


Figure 48: Student Seat Map

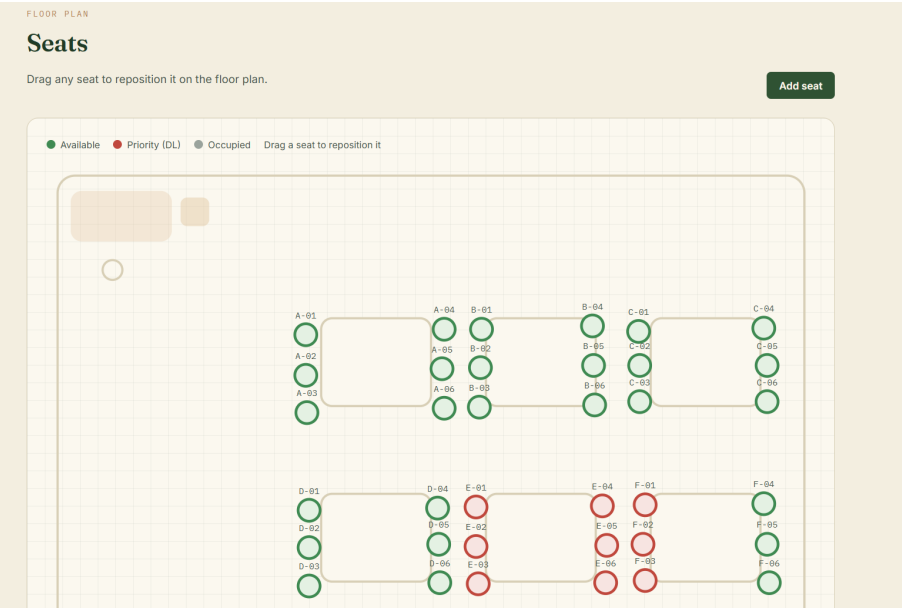


Figure 49: Administrator Seat Map

10.3.2 Reservation Management

The Reservation module allows students to reserve study seats and view their reservation information. On the administrator side, reservations can be monitored and managed according to the system requirements.

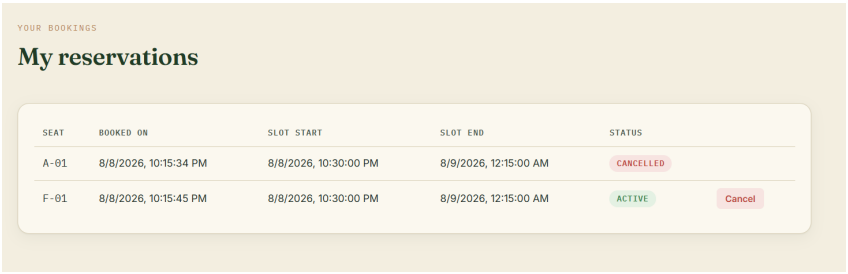


Figure 50: Student Reservation Page

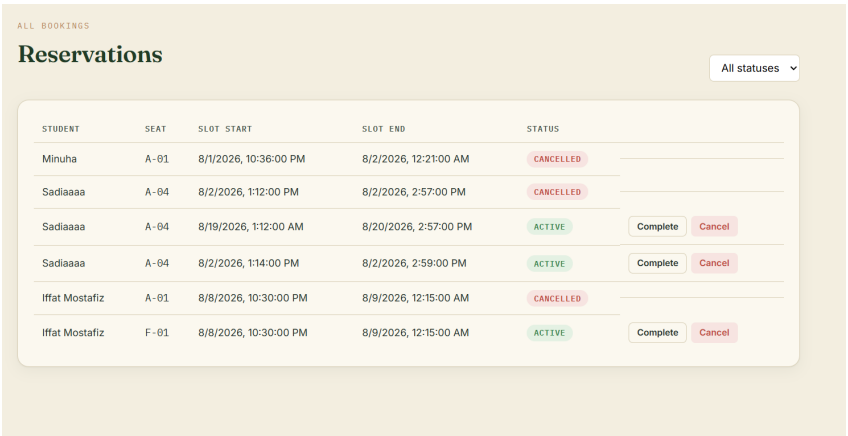


Figure 51: Administrator Reservation Management

10.3.3 Library Management

The Library module provides students with access to library and book information. Students can view the available library resources, while administrators can manage the corresponding library information.

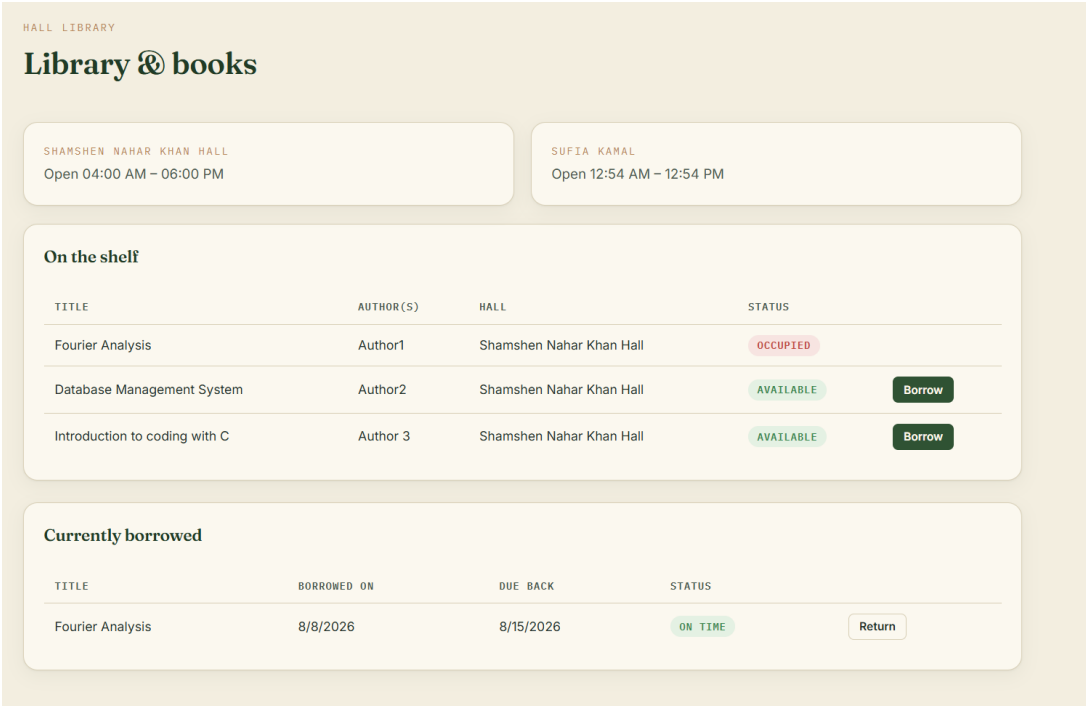


Figure 52: Student Library Page

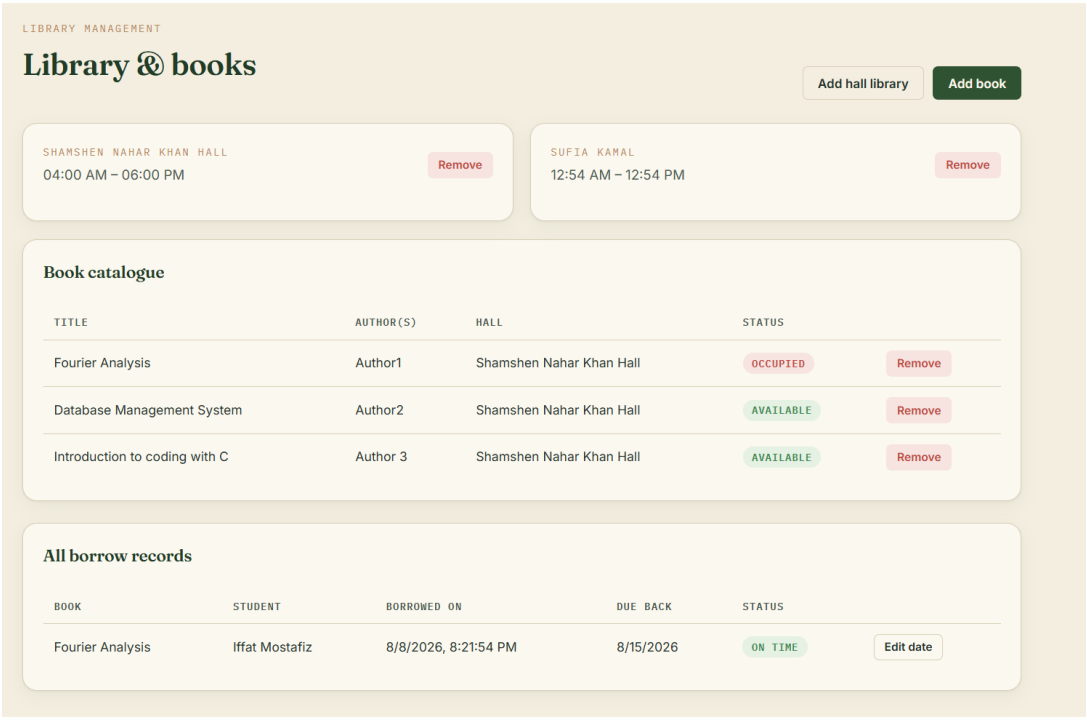


Figure 53: Administrator Library Management

10.3.4 Complaint Management

The Complaint module allows students to submit complaints through the system. Administrators can view and manage the submitted complaints from the administrative interface.

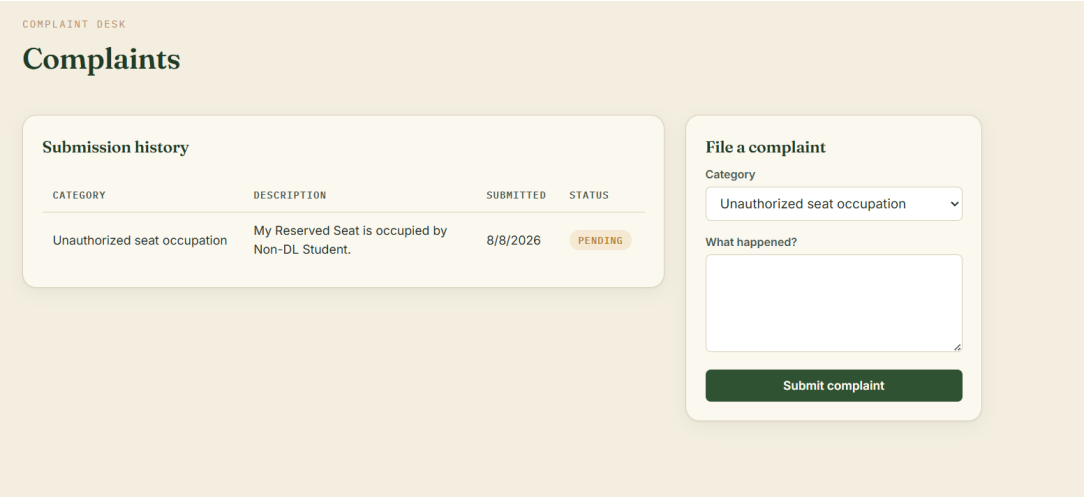


Figure 54: Student Complaint Page

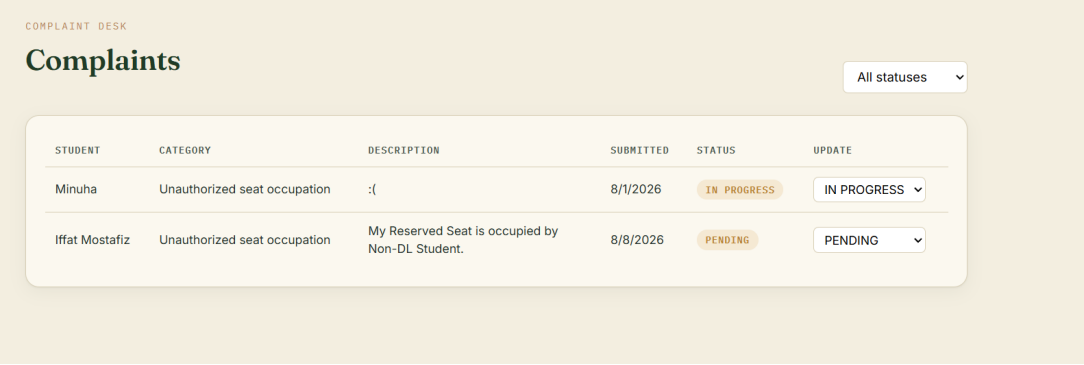


Figure 55: Administrator Complaint Management

10.4 Administrator Management

The Administrator Management section provides administrators with tools to manage important records of the StudyTime system. The management interface includes student management and seat management.

10.4.1 Student Management

The Student Management page allows administrators to view and manage student records stored in the StudyTime system. It provides an interface for managing student-related information.

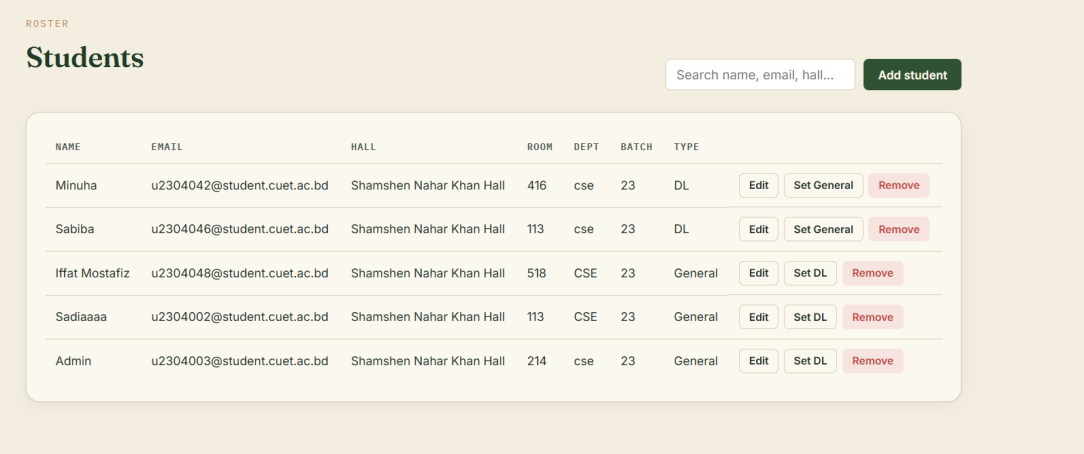


Figure 56: Administrator Student Management Page

10.4.2 Seat Management

The Seat Management page allows administrators to manage the study seats in the system. Administrators can view the available seat information and manage the seats according to the requirements of the system.

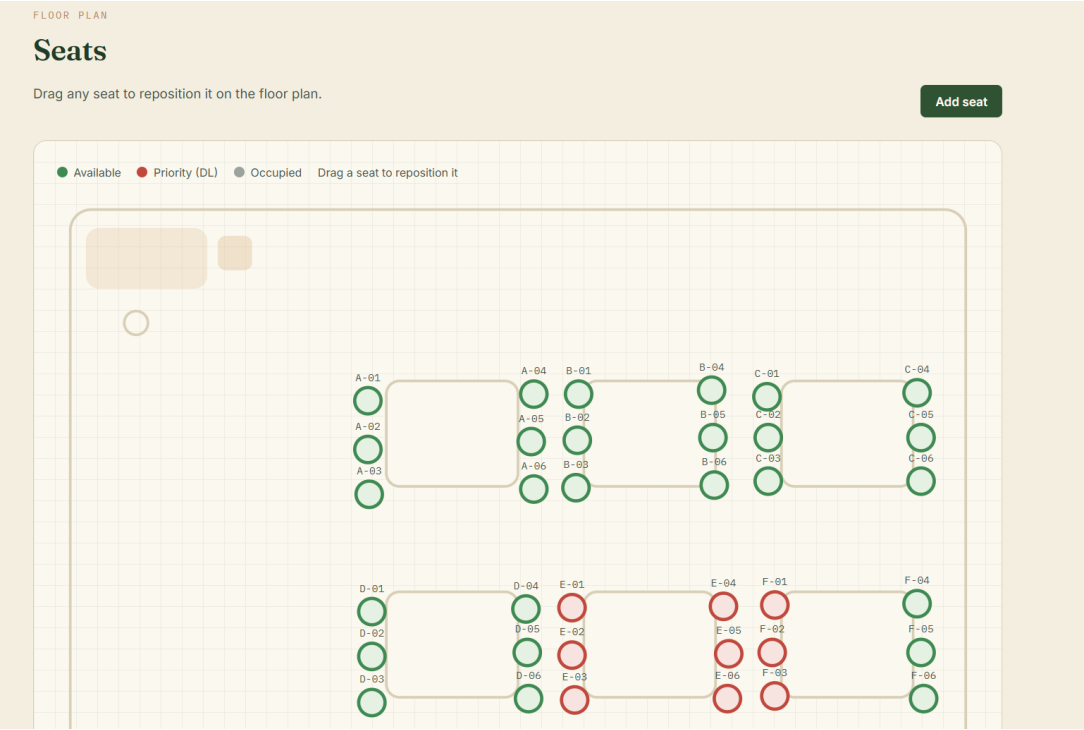


Figure 57: Administrator Seat Management Page

10.5 Implementation Summary

The implemented StudyTime system provides separate interfaces for students and administrators while connecting both sides to the same database structure. Students can register, log in, view the dashboard, select seats, make reservations, access library resources, and submit complaints. Administrators can manage students, seats, reservations, library

information, and complaints. These interfaces demonstrate the practical implementation of the database design and its associated functionalities in a working web application.

11 Future Improvements

The StudyTime system can be further enhanced in several ways:

- Real-time seat availability updates.
- Automated reservation expiry.
- Email or notification support.
- QR-code based seat check-in.
- Advanced library search functionality.
- Automated book-return reminders.
- More detailed database reports.
- Support for multiple university halls.
- Improved administrator analytics.
- Mobile application support.

12 Conclusion

StudyTime demonstrates the practical application of Database Management System concepts through a relational database designed for managing study-room and library facilities.The system contains multiple related relations for students, administrators, seats, reservations, books, libraries, authors, borrowing records, complaints, and relationship management.The project demonstrates important DBMS concepts including primary keys, foreign keys, relational mapping, joins, aggregate functions, grouping, ordering, and CRUD operations.The SQL queries implemented in the project allow useful information to be retrieved from the database, including student reservation history, available seats, library books, borrowing history, complaints, and library statistics.Overall, StudyTime provides a practical example of how database design and SQL can be used to build an organized and efficient management system.